CL-BioGAN: Biologically Inspired Cross-Domain Continual Learning for Hyperspectral Anomaly Detection

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***Abstract*— The memory stability and learning flexibility in continual learning (CL) is a core challenge for cross-scene hyperspectral anomaly detection (HAD) task. Biological neural networks can actively forget history knowledge that conflicts with the learning of new experiences by regulating learning-triggered synaptic expansion and synaptic convergence. Inspired by this phenomenon, we propose a novel biologically inspired CL generative adversarial network (CL-BioGAN) for augmenting continuous distribution fitting ability for cross-domain HAD task, where CL bioinspired loss (CL-Bio Loss) and self-attention generative adversarial network (BioGAN) are incorporated to realize forgetting history knowledge as well as involving replay strategy in the proposed BioGAN. Specifically, a novel bioinspired loss composed with an active forgetting loss (AF Loss) and a CL loss is designed to realize parameters releasing and enhancing between new task and history tasks from a Bayesian perspective. Meanwhile, BioGAN loss with L_2-norm enhances self-attention (SA) to further balance the stability and flexibility for better fitting background distribution for open-scenario HAD (OHAD) tasks. Experimental results underscore that the proposed CL-BioGAN can achieve more robust and satisfying accuracy for cross-domain HAD with fewer parameters and computational cost. This dual contribution not only elevates CL performance but also offers new insights into neural adaptation mechanisms in OHAD task.**

***Index Terms*— Active forgetting, continual learning (CL), generative adversarial network (GAN), hyperspectral anomaly detection (HAD).**

I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds or even thousands of contiguous narrow spectral bands, which makes possibilities for precisely distinguishing different materials [1]. As one of important research fields of hyperspectral information processing, the main aim of open-scenario HAD (OHAD) task is to find pixels that are significantly different from the background in terms of spectral signatures without any prior knowledge of target [2]. Recent years, various traditional deep learning (DL)-based algorithms have been proposed for OHAD tasks [3]. However, most existing DL-based methods mainly excel at acquiring knowledge through generalized learning behavior based on solving specific scene task from a distinct training phase [4]. As shown in Fig. 1(a), traditional DL-based algorithms can only excel at specific task or current scenario because of the specific constructed parameters of network is incapable of dealing with new tasks or scenarios, which is liable to result in catastrophic forgetting phenomenon. Whereas the network parameters of CL can be automatically updated through customized loss functions or automatically updated exemplar set. After the training for t different scenarios, the learned parameters of the model contains the ability of OHAD for all the previous tasks in open scenario circumstance, the procedure is briefly illustrated in Fig. 1(b).

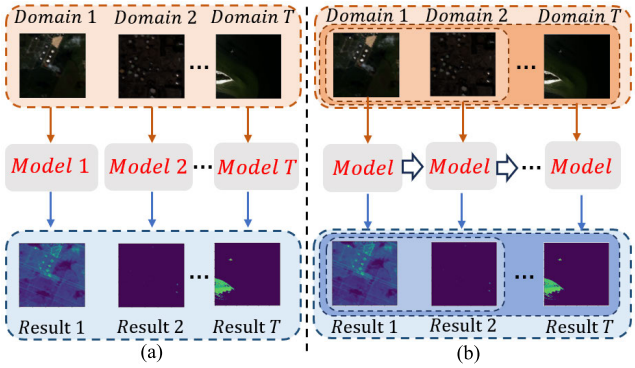


Fig. 1. Comparison of traditional DL training model and the CL method. (a) Traditional DL method, which obtains OHAD results by a set of independent well-trained parameters. (b) Proposed CL method. The parameters of the model are continuously updated with the arrived tasks, but the updated parameters will not forget the previous learned knowledge.

Continual learning (CL) [5] (also known as lifelong learning [6] or incremental learning [7]) as a representative method for alleviating catastrophic forgetting phenomenon caused by varied conditions and scenarios [8], is gradually attracting more and more attention in recent years [9]. Aim at learning knowledge and feature properties of current task while still

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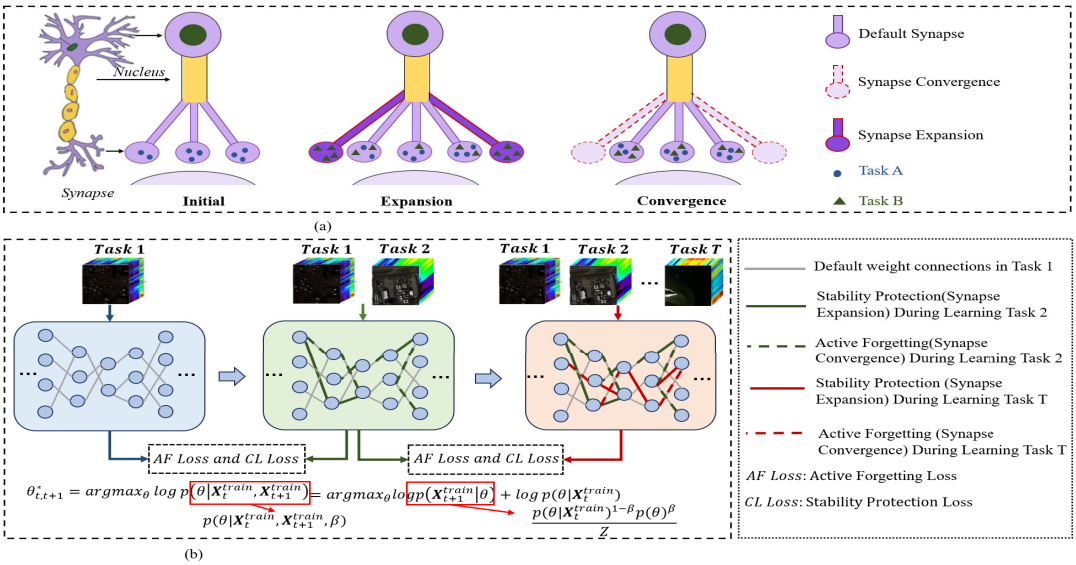


Fig. 2. Principles of active forgetting and learning in our proposed neural network. (a) Neural and protrusion function diagram. (b) Synaptic changes diagram during the learning of a new task.

maintaining the knowledge information from previous training tasks, current methodologies in CL can be conceptually grouped into three main categories: 1) adding regularization terms with reference to the old model (regularization-based approach); 2) approximating and recovering the old data distributions (replay-based approach); and 3) constructing task-specific parameters with a properly designed architecture (architecture-based approach) [10], which have presented proficiencies in addressing numerous CL challenges. However, the effectiveness of strategies varied significantly across different experimental settings, such as task types and similarities, input sizes, and training sample quantities [11]. Therefore, a significant gap exists between research advancements and practical applications especially for OHAD tasks.

Biological neural networks can effectively learn new knowledge based on memorizing history knowledge, even if they conflict with each other. This advantage of memory flexibility is achieved by actively forgetting history knowledge that interferes with the learning of new experiences [12]. Inspired by this biological properties, in order to alleviate catastrophic forgetting problem caused by traditional DL-based method, we explore CL from a biological learning perspective. As shown in Fig. 2(a), the biological active forgetting properties can regulate the learning-triggered synaptic expansion and synaptic convergence [13]. Specifically, biological synapses can expand with additional functional connections to learn new experiences with previously learned functional connections (synaptic expansion), and then prune excrescent connections by transferring useful knowledge to the previously learned synaptic connections (synaptic pruning). Driven by this phenomenon, the proposed biologically inspired CL generative adversarial network (CL-BioGAN) can actively forget history knowledge which interferes with new learning tasks without significantly arousing catastrophic forgetting. We formulate this process with the framework of Bayesian learning to well synthesize and model biological synaptic plasticity by tracking the probability distribution of synaptic weights under dynamic sensory inputs [14], [15]. Specifically, we employ the Bayesian CL incorporate SA to actively forget the posterior distribution, absorbing all information from the history tasks with a forgetting factor to better learn each new task for facilitating CL performance, which fully utilizes bioinspired loss and replay strategy to mitigate the issue of catastrophic forgetting. In addition, the potential spatial feature extraction capability of $\boldsymbol{L}_2$-norm enhanced SA and the local feature extraction capability of CNN are designed through GAN structure to reconstruct the background pixels.

The main innovations and contributions of the proposed CL-BioGAN are summarized as follows.

1) A novel CL-BioGAN is proposed to elaborately integrating CL bioinspired loss (CL-Bio Loss) and BioGAN structure, which can adaptively efficient releasing ineffective parameters as well as strengthening robust parameters through local–global Transformer-based SA learning. CL-BioGAN can provide a dual stable and flexible way from bioinspired network structure and bio-inspired loss to achieve high accuracy with robust performance for cross-domain OHAD task.
2) The proposed BioGAN model is incorporated with a $\boldsymbol{L}_2$ local–global SA discriminator structure, which can efficiently approximate the background distribution through alternating training with bioinspired encoder–decoder generator to acquire local–global representative background spectral characteristics.
3) A novel bioinspired loss is designed to integrate BioGAN loss, active forgetting loss (AF Loss) and CL loss for balancing the memory stability and learning flexibility for OHAD. The BioGAN loss updates the parameters of network for current task and enhances the robustness of BioGAN. AF Loss is formulated with the framework of Bayesian learning, and encourages the model to improve the probability of learning each new task through attenuating old memories by a designed forgetting rate. Meanwhile, the CL loss can automatically balance and maintain the stability of parameters in learning procedure.

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The proposed CL-BioGAN can realize efficient generation and fitting ability to the varied background distribution, which presents satisfying detection accuracy and robust performance for OHAD, and offers a novel approach to the application of DL for open scenarios remote sensing area.

The rest of this article is organized as follows. The related work is introduced in Section II. The detail of the proposed CL-BioGAN framework is introduced in Section III. The related experiments are analyzed and presented in Section IV. The conclusion and future work are presented in Section V.

II. Related Works

A. Continual Learning

Modern machine learning often outperforms human intelligence when it comes to developing robust models from fixed datasets and fixed surroundings [16]. Traditional DL techniques usually suffer from catastrophic forgetting phenomenon for forgetting what has been previously learned by overriding with new data [17]. Therefore, CL methods address this issue by striking a balance between the stability (ability to retain knowledge) and plasticity (ability to learn new concepts) of neural network models [18]. For DL-based CL, similar to the learning of human, the learner typically encounters a nonstationary data stream through a series of learning episodes [19].

Numerous efforts have been devoted to preserving memory stability to mitigate catastrophic forgetting in artificial neural networks. The realization of existing CL methods can be roughly categorized into five mainstreams.

1) *Replay-Based Techniques:* These approaches typically create a replay pool in which features or samples from earlier tasks are maintained to avoid catastrophic forgetting. Generative models is proposed to generate pseudo data for replay strategy [20], [21]. Cheng et al. [22] select the augmentation stability rehearsal (ASR) method by estimating the augmentation stability to select the most representative and discriminative samples. Constructive noises is introduced in each stage of constructive vision transformer (ViT) and an exponential moving average of the working modal is utilized to enforce consistency in predictions [23]. An exemplar-based CL method is introduced by using contrastive learning to learn a task-agnostic and continuously improved feature expression [24]. Rehearsal-based methods aim at handling "stability–plasticity" dilemma by maintaining an appropriate balance between learning new information and retaining previous knowledge [25].
2) *Regularization-Based Techniques:* These techniques are mainly based on minimizing the catastrophic forgetting through utilizing the regularization loss term to minimize the model's parameter distance in various tasks. A dual distillation training objectives is proposed to achieve the combination of two independent models of old classes and new classes [26]. Class similarity knowledge distillation (CSW-KD) method is proposed to perform semantic segmentation by extracting the knowledge of previous models of old classes which are similar to the new class [27]. Three contrastive learning losses are designed to significantly improve the CL effect in the field of remote sensing images [28]. In [29], the elastic weight-consolidated loss function is combined with a self-supervised network. Focusing on the CL model to retain old knowledge while ensuring the model's ability to learn new knowledge, Kim et al. [30] proposed auxiliary network CL (ANCL). Regularization-based methods can effectively avoid the storage of old tasks, therefore it can avoid data leakage and the need for increased memory.
3) *Architecture-Based Methods:* In order to reduce forgetting performance, these methods mainly involve new task-specific layers or structures for new tasks and activities. Parameter masking [31] can learn binary masks on existing networks to obtain a suitable neural network for multiple tasks. Li et al. [32] proposed a method to use architecture search for the best structure of each task. Meta-attention (MEAT) mechanism is designed to determine which image patches and parameters are needed to be isolated by optimizing the binary mask [33]. In [34], the network structure is divided into task-sharing components and task-specific components, where task-specific components can be continuously expanded for new coming tasks. This type of method is adept at dealing with increasing domain shift tasks. In addition, a rule-based model ensemble [35] is introduced by forming a distributed architecture to manage high-dimensional data, where an adjustable weight vector is emphasized by the role of correlation strategies in incremental learning.
4) *Optimization-Based Techniques:* Optimization-based CL algorithms can be achieved by explicitly designing and manipulating the optimization programs. A typical idea is to perform gradient projection. For instance, AdamNSCL [36] instead projects candidate parameter updates into the current null space approximated by the uncentered feature covariance of the old tasks, while AdNS [37] considers the shared part of the previous and the current null spaces. Another attractive idea is meta-learning or learning-to-learn for CL, which attempts to obtain a data-driven inductive bias for various scenarios, rather than designing it manually [16]. ARI [38] combines adversarial attacks with experience replay to obtain task-specific models, which are then fused together through metatraining. MARK [39] maintains a set of shared weights that are incrementally updated with metalearning and selectively masked to solve specific tasks. Besides, some other works refine the optimization process from a loss landscape perspective. Linear connector [40] adopts Adam-NSCL [36] and feature distillation to obtain respective solutions of the old and new tasks connected by a linear path of low error, followed by linear averaging.
5) *Representation-Based Techniques:* Have attempted to incorporate the advantages of self-supervised learning (SSL) and large-scale pretraining to improve the representations in initialization and in CL. Based

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on SSL strategy (basically with contrastive loss) for CL, MinRed [41] further promotes the diversity of experience replay by decorrelating the stored old training samples. CaSSLe [42] converts the self-supervised loss to a distillation strategy by mapping the current state of a representation to its previous state. Based on pretraining for downstream CL, representative strategies include selecting the most relevant prompts from a prompt pool (L2P [43]), optimizing a weighted summation of the prompt pool with attention factors (CODAPrompt [44]), using explicitly task-sharing and task-specific prompts (DualPrompt [45]) or only task-specific prompts (SPrompts [46]), and progressive expansion of task-specific prompts (progressive prompts [47]).

B. Hyperspectral Anomaly Detection

HAD task aims at detecting the anomalies from background samples through the significant difference between continuous spectral signatures [48]. Existing HAD algorithms can be roughly divided into three mainstreams.

1) *Statistic-Based HAD Approaches:* Mainly estimate the probability of anomalous samples primarily by calculating the relationship difference between the background and the anomalous distribution by calculating differences. Reed–Xiaoli (RX) [48] is proposed to detect anomalies by computing the mean and variance of the entire image through comparing the Mahalanobis distance between the test sample and the background mean vector. Many variations based on RX, such as kernel-based RX (KRX) algorithm [49], local RX (LRX) [50] have been gradually proposed. Meanwhile, a tensor RX (TRX) algorithm based on fractional Fourier transform [51] and the recursive RX with extended multi-ttribute profiles [52] are designed to extract more image feature information. In [53], an adaptive reference correlation graph embedding (ARGE) is proposed to effectively obtain low-dimensional features and improve computational efficiency. To overcome the limitations of distributional assumptions in HAD, recent years have seen increased interest in.
2) *Representation-Based HAD Approaches:* The cooperative representation-based detector (CRD) [54] assumes that background pixels can be minimized by enhancing the $\boldsymbol{L}_2$-norm of the representation weight vector, approximately represented by a linear combination of its spatial neighbors. LRASR [55] assumes that backgrounds present low-rank attributes and anomalies present sparse attributes. Chang [56] proposed an effective anomaly space (EAS) to solve the problem of anomalies being sandwiched between the background and noise during the background suppression (BS) process in HAD. The total variation (TV) representation model is introduced into HAD [57], such as the new enhanced TV (ETV) of the endmember background dictionary (EBD) to represent the row vector of the coefficient matrix [58]. Feedback band group and variation low-rank sparse model (FBGVLRS-AD) is proposed by utilizing local spatial constraints within spectral features combined with the TV tensor-based HAD method [59].
3) *DL-Based HAD Approaches:* Can learn complex data distribution even under the condition of lack of prior information [60]; therefore, background estimation and reconstruction are key issues in the implementation of HAD. Based on the ability for hierarchical learning, abstraction, and high-level representation of autoencoder (AE), which provides an unsupervised manner for background reconstruction [61]. The background-guided deformable convolutional AE (DCAE) network is designed for HAD [62]. In [63], a weighted adaptive loss combined with AE is designed to suppress the reconstruction of abnormal samples. PASSNet [47] built a lightweight hybrid model, which ensembles the respective inductive bias from the convolutional neural network (CNN) and global receptive field from transformers with a designed patch attention module (PAM) to extract spatial–spectral features from multiple spatial perspectives by integrating both CNNs and transformer blocks. In terms of feature representation capabilities and adversarial training capabilities, generative adversarial network (GAN) has been successfully used to estimate background distribution and spectral domain features to achieve high-confidence background sample reconstruction [64], [65]. Under the condition that the number of background samples is much larger than abnormal samples, the recognition error of abnormal pixels is higher than that of background pixels. Jiang et al. [66] only used the normal background samples to train the GAN to estimate the background distribution to achieve the detection of abnormal samples. Based on the visual attention Transformer model, a clustering module is introduced to detect false background and abnormal samples in [67]. In [68], an HAD network based on variational background inference and generative adversarial framework (VBIGAN-AD) is established based on the relationship between data samples and latent samples through two subnetworks to capture the data distribution.

C. Bayesian CL

CL needs to remember the old tasks and learn each new task effectively. Therefore, considering a neural network with parameter θ continually learns two independent task t and task $t+1$ based on their training datasets $\mathbf{X}_t^{\text{train}}$ and $\mathbf{X}_{t+1}^{\text{train}}$. All the training datasets for each task is only available when learning the task; therefore, the posterior distribution and the Bayesian CL can be represented as follows.

1) Bayesian Learning: After learning $\mathbf{X}_t^{\text{train}}$, the posterior distribution

$$p\left(\theta|\mathbf{X}_t^{\text{train}}\right) = \frac{p\left(\mathbf{X}_t^{\text{train}}|\theta\right)p(\theta)}{p\left(\mathbf{X}_t^{\text{train}}\right)} \tag{1}$$

incorporates the knowledge of task t. Then, we can get the predictive distribution for the testing data of task t

$$p\left(\mathbf{X}_t^{\text{test}}|\mathbf{X}_t^{\text{train}}\right) = \int p\left(\mathbf{X}_t^{\text{test}}|\theta\right)p\left(\theta|\mathbf{X}_t^{\text{train}}\right)d\theta \tag{2}$$

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whereas the posterior $p(\theta|\mathbf{X}_t^{\text{train}})$ is generally intractable (except very special cases), we must resort to approximation methods, such as the Laplace approximation or other approaches of approximate inference [69]. Let us take Laplace approximation as an example. If $p(\theta|\mathbf{X}_t^{\text{train}})$ is smooth and majorly peaked around the mode $\theta_t^* = \arg\max_\theta \log p(\theta|\mathbf{X}_t^{\text{train}})$, we can approximate it with a Gaussian distribution whose mean is θ_t^* and covariance is the inverse Hessian of the negative log posterior [13].

2) Bayesian CL: Further for incorporating the new task into the posterior, the posterior $p(\theta|\mathbf{X}_t^{\text{train}})$ is used as the prior of the next task

$$p\left(\theta|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}\right) = \frac{p\left(\mathbf{X}_{t+1}^{\text{train}}|\theta\right) p\left(\theta|\mathbf{X}_t^{\text{train}}\right)}{p\left(\mathbf{X}_{t+1}^{\text{train}}\right)}. \tag{3}$$

Then, we can test the performance of CL by evaluating

$$p\left(\mathbf{X}_t^{\text{test}}, \mathbf{X}_{t+1}^{\text{test}}|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}\right) = \int p\left(\mathbf{X}_t^{\text{test}}, \mathbf{X}_{t+1}^{\text{test}}|\theta\right) p\left(\theta|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}\right) d\theta. \tag{4}$$

Similarly, $p(\theta|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}})$ can be approximated by a Gaussian using Laplace approximation whose mean is the mode of the posterior

$$\begin{aligned}\theta_{t,t+1}^* &= \arg\max_\theta \log p\left(\theta|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}\right) \\ &= \arg\max_\theta \log p\left(\mathbf{X}_{t+1}^{\text{train}}|\theta\right) + \log p\left(\mathbf{X}_t^{\text{train}}|\theta\right) \\ &\quad - \underbrace{\log p\left(\mathbf{X}_{t+1}^{\text{train}}\right)}_{\text{const.}}.\end{aligned} \tag{5}$$

III. Method

In this section, we mainly introduce the proposed CL-BioGAN by combining knowledge learned from history and present tasks. The overall architecture of the proposed CL-BioGAN is represented in Fig. 3, which is mainly composed of the CL structure and BioGAN networks.

Suppose that there are T different scenario domains $\mathbf{X}_1, \ldots, \mathbf{X}_t, \ldots, \mathbf{X}_T$ for OHAD task datasets, each $\mathbf{X}_t$ represents the tth HAD scenario task. An HSI can be represented as $\mathbf{X}_t \in \mathbb{R}^{M\times N\times C} = \{\mathbf{x}_i \in \mathbb{R}^C\}_{i=1}^{i=M\times N}, t \in 1, 2, \ldots, T$, where $\mathbf{x}_i$ represents the ith spectral vector in $\mathbf{X}_t$.

A. Structure of BioGAN

The whole BioGAN for OHAD mainly consists of three components: the background selection module (BSM), the generator module G, and the discriminator module D.

1) Background Selection Module: BSM is shown in Fig. 3(b), which consists of the two main processes: 1) spectral-based background selection strategy (SBSS) and 2) spatial–spectral noise smoothing strategy (SSNS). The pseudolabel for the background samples and anomalies are first coarsely selected through SBSS. Then, SSNS is used to involve spatial information for better exploiting and enhancing spectral–spatial features for each vector.

a) Spectral-based background selection strategy: Aiming to approximately obtain the background datasets, we calculate the similarity value of neighboring spectral vectors based on SAM [70] for coarsely selecting the background sample sets $\mathbf{B}_t$ for any HAD scenario $\mathbf{X}_t$

$$\text{SAM}\left(\mathbf{x}_i, \mathbf{x}_j\right) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{||\mathbf{x}_i|| \times \left|\left|\mathbf{x}_j\right|\right|} \tag{6}$$

$$\mathbf{s}_i = \begin{cases} 0, & \text{SAM}\left(\mathbf{x}_i, \mathbf{x}_j\right) \geq \mu \\ 1, & \text{SAM}\left(\mathbf{x}_i, \mathbf{x}_j\right) < \mu \end{cases} \tag{7}$$

$$\mathbf{B}_t = \left\{\mathbf{index}_{(i)}\{\mathbf{s}_i == 0\}\right\}. \tag{8}$$

The threshold μ is defined to determine whether $\mathbf{x}_i$ is a background sample or an anomalous sample. The value of $\mathbf{s}_i = 0$ indicates that the current pixel is part of the background, where $\mathbf{s}_i = 1$ indicates the current pixel is abnormal pixel, where $\mathbf{B}_t \in \mathbb{R}^{n_b\times C}$ and n_b represents the number of background.

b) Spatial–spectral noise smoothing strategy: We developed SSNS to incorporate spatial pixel information into spectral data, which concatenate the original spectral vector values with the average value of the spectral vectors of the $w \times w$ local region around the center pixel (in the experiment, we take w as 3), and compute the mean vector as follows:

$$\bar{\mathbf{x}}_i = \frac{1}{w \times w} \sum_{r=1}^{w^2} \mathbf{x}_i^r \tag{9}$$

where r is the pixel's index in the $w \times w$ region. The original spectral vector $\mathbf{x}_i$ and the mean vector $\bar{\mathbf{x}}_i$ are merged to create the enhanced vector $\mathbf{e}_i$

$$\mathbf{e}_i = \mathbf{x}_i \otimes \bar{\mathbf{x}}_i \tag{10}$$

where $\mathbf{e}_i \in \mathbb{R}^{2C}$ and $\otimes$ represents the concatenating operation across the channel dimension. Therefore, the augmented spectral–spatial background sample sets $\mathbf{B}_t$ updated as follows:

$$\mathbf{B}_t \in \mathbb{R}^{n_b\times 2C} = \left\{\mathbf{e}_i |\ \mathbf{index}_{(i)}\{\mathbf{s}_i == 0\}\right\}. \tag{11}$$

2) Generator Module G: Further for improving reconstruction performance of background pixels, we introduce an advanced encoder–decoder architecture, where the structure of G is shown in Fig. 3(b). The AE [71] has been widely used in various applications such as HSI unmixing [72] and target detection [73] because of the simple and outstanding performance in feature extraction. For the simplest AE structure, in which the encoder and decoder are both fully connected layers. Therefore, for the arbitrary selected background input $\mathbf{b}_i \subset \mathbf{B}_t \in \mathbb{R}^{n_b\times 2C}$ and the hidden layer $\mathbf{z}_i \subset \mathbf{Z}_t \in \mathbb{R}^{n_b\times L}$ with $L(L < 2C)$ dimensions, the reconstructed output $\hat{\mathbf{b}}_i \subset \hat{\mathbf{B}}_t \in \mathbb{R}^{n_b\times 2C}$ can be calculated as follows:

$$\mathbf{z}_i = f(\mathbf{W}_{\text{en}}\mathbf{b}_i + \mathbf{bias}_{\text{en}}) \tag{12}$$

$$\hat{\mathbf{b}}_i = f(\mathbf{W}_{\text{de}}\mathbf{z}_i + \mathbf{bias}_{\text{de}}) \tag{13}$$

where $\mathbf{W}_{\text{en}}$ and $\mathbf{bias}_{\text{en}}$ and $\mathbf{W}_{\text{de}}$ and $\mathbf{bias}_{\text{de}}$ denote the weights and biases of the encoder and decoder, respectively, and $f(\cdot)$ is an inconclusive activation function. Based on the reality that the majority of pixels in HSIs belong to the background

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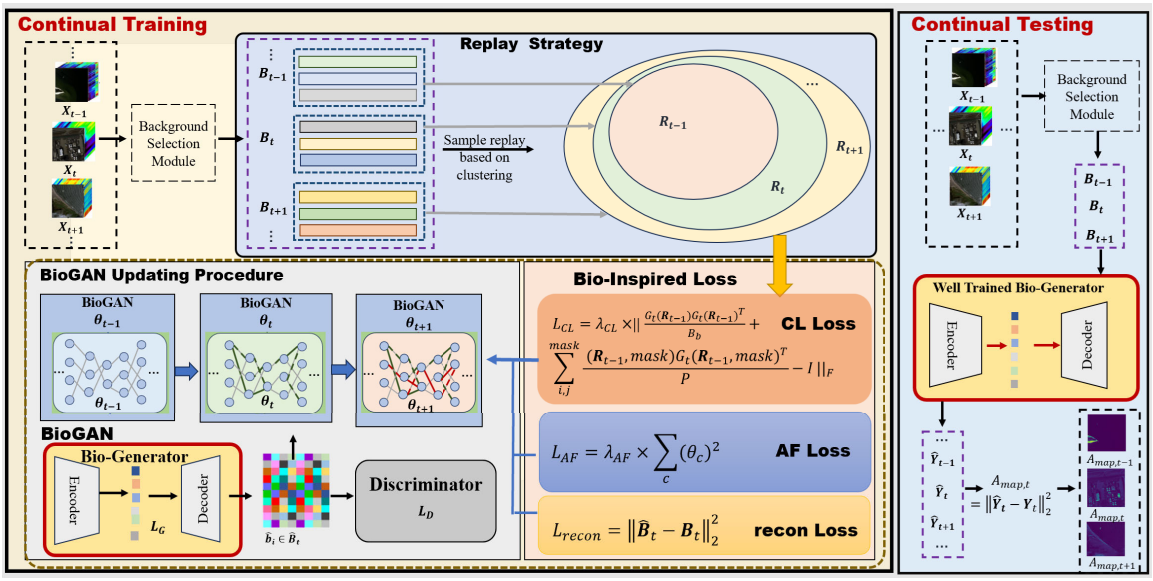


Fig. 3. Overview of the proposed CL-BioGAN for OHAD. The left part represents the continual training phase, which is mainly composed of replay strategy, BioGAN structure, and bioinspired loss (CL Loss, AF Loss, and recon Loss). The right part in the figure is the continual testing part, the well-trained biogenerator is only used in the testing phase to detect the continually arrived HAD scenarios.

while only a few pixels are anomalies, AE mainly progressively emphasizes the background features during the training process, and finally reconstructed background pixels as the inputs background pixels as similar as possible. Therefore, the training goal of the proposed BioGAN is to learn the features of the training dataset $\mathbf{b}_i$ and reconstruct $\hat{\mathbf{b}}_i$ as similar as possible. The reconstruction error of a well-trained AE can be used to calculate the final anomaly detection result.

3) Discriminator Module D: D is shown in Fig. 3(b), which is used to learn more complicated spectral properties as well as support the training of G. D consists of two modules: the multi-scale convolution (MSC) module for extracting multiscale features, and $\boldsymbol{L}_2$ SA [$\boldsymbol{L}_2$ norm self-attention (SA)] module for channel feature extraction and channel correlation. Three multiscale spectral feature maps $\boldsymbol{F}$ have been generated by using 1×1, 3×3, and 5×5 convolutional operations. Meanwhile, residual connection structure is utilized in $\boldsymbol{L}_2$ SA module, and feature information in different levels can be extracted by concatenating the feature maps of several receptive fields together, and the spatial position embedding is introduced to capture the local discrimination details and the general characteristic changes of the spectral information. For an arbitrary input $\hat{\mathbf{B}}_t \in \mathbb{R}^{n_b \times 2C}$, the corresponding output of an SA block can be formulated as follows:

$$\begin{aligned} \mathbf{Q} &= \mathbf{W}_q \boldsymbol{F} \in \mathbb{R}^{n_b \times L} \\ \mathbf{K} &= \mathbf{W}_k \boldsymbol{F} \in \mathbb{R}^{n_b \times L} \\ \mathbf{V} &= \mathbf{W}_v \boldsymbol{F} \in \mathbb{R}^{n_b \times L} \end{aligned} \tag{14}$$

$$\mathbf{SA} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{||\mathbf{Q}||_2 ||\mathbf{K}||_2^T}{\sqrt{2C}}\right)\mathbf{V} \tag{15}$$

where $\mathbf{W}_q \in \mathbb{R}^{2C \times L}$, $\mathbf{W}_k \in \mathbb{R}^{2C \times L}$, and $\mathbf{W}_v \in \mathbb{R}^{2C \times L}$ are three predefined transformation matrices, respectively. $||\cdot||_2$ represents the $\boldsymbol{L}_2$ norm of $\mathbf{SA} \in \mathbb{R}^{n_b \times L}$. We can observe from (14) and (15) that the $\mathbf{SA}$ considers the relationships between each input and all other inputs. Especially for HSIs, the $\mathbf{SA}$ can fully excavate more important spatial information for background reconstruction.

A linear full-connection layer is utilized for converting the intermediate feature dimension to the output feature dimension $2C$, after the completion of the $\boldsymbol{L}_2$ SA computation. The following final attention feature map is calculated as follows:

$$\hat{\boldsymbol{F}} = \text{FFN}((\text{Linear}(\mathbf{SA}) + \boldsymbol{F})) + (\text{Linear}(\mathbf{SA}) + \boldsymbol{F}) \tag{16}$$

where $\hat{\boldsymbol{F}} \in \mathbb{R}^{n_b \times 2C}$ and FFN represents feedforward neural network, which consists of dropout, linear, and the stable GELU activation function. Then, $\hat{\boldsymbol{F}}$ is imported to the a multiple layer perception (MLP) to obtain the final discrimination result.

B. Sample Replay Based on Clustering

After t task has been learned, in order to alleviate the catastrophic forgetting problem, an alternative sampling approach for sample replay selection is designed by involving sample replay strategy to random choose and store pixels from historical data. Suppose the background set $\mathbf{B}_t$ with totally M_t samples, we cluster the whole dataset into P groups by utilizing K-means technique. Overall, the expectation–maximization (EM) optimization algorithm is used [74], and the optimization objective function is as follows:

$$E = \sum_{i=1}^{P} \sum_{x \in C_i} \|x - \mu_i\|^2 \tag{17}$$

where C_i represents the set of all sample points of each cluster, x denotes a sample point belonging to C_i, and μ_i represents the center of the ith cluster. Each group will contain $M_t^i (i = 1, 2, \ldots, P)$ samples. We assume that each task adaptively selects N representative samples by taking into account memory restriction. We take the sum of one percent of M_t^i closest to the sample center in each cluster, $N = \sum_{i=1}^{P} (M_t^i \times 100)/10\,000$ denoted by N. The specific

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value of N is detailed in Section IV. The sample replay equation can be represented as follows:

$$k(\mathbf{B}_t) = \bigcup_{i=1}^{P} K_{\text{Means}}\left[:\left\lfloor \frac{N * M_t^i}{M_t} \right\rfloor\right] \tag{18}$$

where samples in K_{Means} have been organized in ascending order based on their distance from each cluster's center pixel, and $((N * M_t^i)/M_t)$ is the picked data from each cluster sets. The computed representative sample set of the ith cluster is represented by the symbol $K_{\text{Means}}[: \lfloor (N * M_t^i)/M_t \rfloor]$. The replay buffer may be modified as follows:

$$\mathbf{R}_t \leftarrow \mathbf{R}_{t-1} \cup k(\mathbf{B}_t), \quad \mathbf{R}_{t-1} = \phi, \text{ when } t = 1. \tag{19}$$

We may lessen the effects of the imbalanced data distribution and increase the spectral anomaly detection algorithm's resilience by using (19) for adaptive replay technique. Obviously, abnormalities in various HSI sets may be continually detected with better accuracy performance based on the clustering sample replay in CL module.

C. CL-Bio Loss Function

In this section, we mainly introduce the proposed bioinspired loss of CL based on the Bayesian learning framework, which hypothesized based on biological synaptic plasticity by tracking the probability distribution of synaptic weights under dynamic sensory inputs [14]. The whole proposed CL-bio loss function is composed with: 1) ACL loss L_{ACL} and 2) the BioGAN loss L_{BioGAN}. The ACL loss is used to assure stability for detection and prevent catastrophic forgetting for cross-domain scenarios. BioGAN loss mainly ensures the training fidelity performance for background sample reconstruction. Therefore, we construct and summarize the whole loss function as follows:

$$L = L_{\text{BioGAN}} + L_{\text{ACL}} \tag{20}$$

$$L_{\text{BioGAN}} = L_G + L_D + L_{\text{recon}} \tag{21}$$

$$L_{\text{ACL}} = L_{\text{CL}} + L_{\text{AF}}. \tag{22}$$

1) BioGAN Loss: The spatial–spectral background sets $\mathbf{B}_t$ and reconstruction background set $\hat{\mathbf{B}}_t = G(\mathbf{B}_t)$ is measured by using the mean square error (mse) loss

$$L_{\text{recon}} = \left\| \hat{\mathbf{B}}_t - \mathbf{B}_t \right\|_2^2. \tag{23}$$

We performed differentiable data enhancement (Diff) on the data and pseudodata in the G and D to assure the training stability of GAN without appearing feature shift phenomenon for brightness, saturation, and contrast information to realize better convergence results. Consequently, the adversarial process allows G and D to be updated alternately as follows:

$$L_G = \mathrm{E}_{\mathbf{x}\sim p(\mathbf{B}_t)}\left[\log(1 - D(\text{Diff}(G(\mathbf{x}))))\right] \tag{24}$$

$$L_D = -E_{\mathbf{x}\sim p(\mathbf{B}_t \cup \hat{\mathbf{B}}_t)}\left[\log(D(\mathbf{x})) + \log(1 - D(\text{Diff}(G(\mathbf{x}))))\right]. \tag{25}$$

2) ACL Loss: ACL loss is composed of CL loss to prevent weight drift caused by new knowledge and the AF Loss for selectively converging the related parameter weights of the network.

a) AF Loss : Inspired by biological forgetting and Bayesian CL, we introduce a forgetting factor β and replace $p(\theta|\mathbf{X}_t^{\text{train}})$ that absorbs all the information of $\mathbf{X}_t^{\text{train}}$ with a weighted product distribution

$$\hat{p}\left(\theta|\mathbf{X}_t^{\text{train}}, \beta\right) = \frac{p\left(\theta|\mathbf{X}_t^{\text{train}}\right)^{(1-\beta)} p(\theta)^{\beta}}{Z} \tag{26}$$

where we denote that $p(\theta|\mathbf{X}_t^{\text{train}})$ and $p(\theta)$ are mixed to produce the new distribution $\hat{p}$. Z is the normalization value that depends on β, which keeps $\hat{p}(\theta|\mathbf{X}_t^{\text{train}}, \beta)$ following a Gaussian distribution if $p(\theta|\mathbf{X}_t^{\text{train}})$ and $p(\theta)$ are both Gaussian. When $p(\theta) \to 0$, $\hat{p}$ will be dominated by $p(\theta|\mathbf{X}_t^{\text{train}})$ and remember all the information about task t, where $p(\theta) \to 1$, $\hat{p}$ will actively forget all the information about task t. By modifying (4), the new target becomes

$$p\left(\mathbf{X}_t^{\text{test}}, \mathbf{X}_{t+1}^{\text{test}}|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}, \beta\right) = \int p\left(\mathbf{X}_t^{\text{test}}, \mathbf{X}_{t+1}^{\text{test}}|\theta\right) p\left(\theta|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}, \beta\right) d\theta \tag{27}$$

where β decides how much information from task t should be forgotten to maximize the probability of learning task $t+1$. A suitable β value should be defined as follows:

$$\begin{aligned}\beta^* &= \arg\max_{\beta} p\left(\mathbf{X}_{t+1}^{\text{train}}|\mathbf{X}_t^{\text{train}}, \beta\right) \\ &= \arg\max_{\beta} \int p\left(\mathbf{X}_{t+1}^{\text{train}}|\theta\right) \hat{p}\left(\theta|\mathbf{X}_t^{\text{train}}, \beta\right) d\theta.\end{aligned} \tag{28}$$

Here, β should be between 0 and 1, we make a grid search to determine the optimal value for β. Next, $p(\theta|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}, \beta)$ can also be estimated by the approximated Gaussian using Laplace approximation, and the maximum a posteriori (MAP) estimation is

$$\begin{aligned}\theta_{t,t+1}^* &= \arg\max_{\theta} \log p\left(\theta|\mathbf{X}_t^{\text{train}}, \mathbf{X}_{t+1}^{\text{train}}, \beta\right) \\ &= \arg\max_{\theta} (1-\beta)\left(\log p\left(\mathbf{X}_{t+1}^{\text{train}}|\theta\right)\right. \\ &\quad \left. + \log p\left(\theta|\mathbf{X}_t^{\text{train}}\right)\right) + \beta \log p\left(\theta|\mathbf{X}_{t+1}^{\text{train}}\right) + \text{const.}\end{aligned} \tag{29}$$

Then, we obtain the active forgetting AF Loss function by synaptic expansion-convergence (AF Loss)

$$L_{\text{AF}} = \lambda_{\text{AF}} \times \sum_c (\theta_c)^2 \tag{30}$$

where θ_c represents the parameters of the current task. Fig. 2(b) indicates the implementation of active forgetting in a CL model.

b) CL loss: This loss function minimizes the magnitude of parameter updates by forcing the distance between the input data of task t and the reconstructed image of task $t+1$ to get closer to preserve stability for OHAD. We use a replay sample set $\mathbf{R}_t$ to reduce the distance between different task models for stability protection in CL. Finally, the definition of the F-norm regularization loss for every task t is

$$L_{\text{CL}} = \lambda_{\text{CL}} \times \left\| \frac{G_t(\mathbf{R}_{t-1}) G_t(\mathbf{R}_{t-1})^T}{B_b} \right.$$

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Algorithm 1 CL-BioGAN

Input: The stream HSIs $\mathbf{X}_1, \ldots, \mathbf{X}_t, \ldots, \mathbf{X}_T$ for T tasks.
Parameter: $\mu = 0.99$, the number of clustering groups $P = 3$, $w = 3$, the channel dimension $C = dmin(\mathbf{X}_1, \ldots, \mathbf{X}_T)$, $\lambda_{AF} = 0.1$, $\lambda_{CL} = 0.9$, $lr = 0.00005$.
Output: $A_{map,t}$.

```
Initialize replay buffer R_1 = φ, the model parameters θ_1.
for t = 1 to T tasks do
    X_t ∈ R^{M×N×C} = {x_i ∈ R^C}_{i=1}^{i=M×N};
    for H training epochs do
        generate s_i vector by (6) and (7);
        construct background samples set B_t by (11);
        if t = 1 then
            update θ_1 by minimizing L_G, L_D and L_recon through
            equation (24), (25) and (23), as equation (30) L_AF = 0,
            equation (31) L_CL = 0;
        else
            update θ_t by minimizing L_G, L_D and L_recon through
            equation (24), (25) and (23);
        end if
        apply K-Means to obtain M_i(i = 1, 2, 3) containing three
        groups exemplars respectively for the subset through (18);
        if t = 1 then
            update replay buffer by R_1 ← k(B_t);
        else
            update replay buffer by R_t ← R_{t−1} ∪ k(B_t);
        end if
    end for
end for
Construct the detection for tasks through equation (20);
```

$$+\sum_{i,j}^{\text{mask}} \frac{G_t(\mathbf{R}_{t-1,\text{mask}})G_t(\mathbf{R}_{t-1,\text{mask}})^T}{P} - \mathbf{I}\Bigg\|_F. \tag{31}$$

Among them, B_b stands for batch size, P denotes the number of pixel in mask. $\mathbf{I}$ stands for the identity matrix, $\|\bullet\|_F$ represents the matrix's Fibonacci norm, and λ_{CL} represents the balancing parameter. The CL loss will gradually update the parameters of the generator in accordance with the continuously coming tasks.

D. Continual Anomaly Detection

During the testing procedure in Fig. 3(c), the final projected anomaly probability map is obtained by subtracting the continuous data $\mathbf{Y}_1, \ldots, \mathbf{Y}_t, \ldots, \mathbf{Y}_T$ from the reconstructed background pixel $\hat{\mathbf{Y}}_1 = G(\mathbf{X}_1), \ldots, \hat{\mathbf{Y}}_t = G(\mathbf{X}_t), \ldots, \hat{\mathbf{Y}}_T = G(\mathbf{X}_T)$ by our proposed well-trained CL-BioGAN

$$A_{\text{map},t} = \left\|\hat{\mathbf{Y}}_t - \mathbf{Y}_t\right\|_2^2. \tag{32}$$

The detailed pseudocode for the proposed CL-BioGAN is introduced in Algorithm 1.

IV. Experiments

In this section, we mainly assess the proposed CL-BioGAN for the open scenario OHAD on five cross domain distinct hyperspectral datasets. In the following section, we will introduce and analyze the performance of the proposed CL-BioGAN from experimental setting, detection results, and ablation experiments under open scenario circumstances.

TABLE I
Details of the ABU Anomaly Detection Dataset

HSIs	Spatial size	channels	resolution/m	bands/nm	Sensor
Los Angeles-1	100×100	205	7.1	430 - 860	AVIRIS
Bay Champagne	100×100	188	4.4	400 - 2500	AVIRIS
Los Angeles-2	100×100	205	7.1	430 - 860	AVIRIS
Cat Island	150×150	188	17.2	400 - 2500	AVIRIS
San Diego	100×100	193	7.5	400 - 2500	AVIRIS

TABLE II
Details of the HAD100 Anomaly Detection Dataset

HSIs	Spatial size	channels	resolution/m	Date	Sensor
HAD100_91	64×64	276	2.3	2017/8/21	AVIRIS-NG
HAD100_100	64×64	276	2.3	2017/8/21	AVIRIS-NG
HAD100_24	64×64	276	8.4	2019/10/4	AVIRIS-NG
HAD100_33	64×64	276	1.9	2021/6/14	AVIRIS-NG
HAD100_40	64×64	276	2.0	2017/9/8	AVIRIS-NG

A. Experimental Setting

1) Datasets and Computing Infrastructure: We evaluated our method on two types of datasets: ABU-Urban and ABU-Beach (ABU) [75], as well as HAD100 for OHAD [76]. For the ABU dataset, we select five different real HSIs to form the CL tasks: Los Angeles-1, Bay Champagne, Los Angeles-2, Cat Island, and San Diego. These HSIs are captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor under different scenarios with distinct spatial resolutions, channels, spatial size, and spectral bands. The detailed information of these datasets is illustrated in Table I. Meanwhile, we randomly select five different areas from the HAD100 dataset as shown in the table as HAD10Data as ang20170821t183707_91, ang20170821t183707_100, ang20191004t185054_24, ang20210614t141018_33, ang20170908t225309_40, are, respectively, denoted by HAD100_91, HAD100_100, HAD100_24, HAD100_33, and HAD100_40, which are all captured by Airborne Visible/Infrared Imaging Spectrometer-next generation (AVIRIS-NG). The details of the HAD100 dataset are shown in Table II. All the experiments conducted under the hardware environment: Intel[1] Xeon[1] CPU at 2.30 GHz with 64-GB RAM and an NVIDIA GeForce GTX 2080 Ti graphical processing unit (GPU) with 11-GB RAM, and a 64-bit Windows 10 system and the Pytorch 1.10.2 DL frameworks.

2) Methods: In order to demonstrate the superiority of our proposed CL-BioGAN for OHAD, we select the most representative methods from each paradigm to establish the benchmark. These training methods include DL-based joint (J) training for OHAD, fine-tune (FT)-based OHAD, and CL-based algorithm MAS [77], EWC [17], OWM [78], and CL-CaGAN [79].

3) Metrics: In this article, we mainly utilize receiver operating characteristics (ROCs) [80] and the area under the ROC curve (AUC) [81] to quantitatively assess the performance of detectors. The ROC curve illustrates the relationship between

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true positive rate (P_D) and false positive rate (P_F). A 3-D ROC curve can be generated by a triplet parameter vector specified by (P_D, P_F, and τ), or by three 2-D ROC curves of (P_D and P_F), (P_D and τ), and (P_F and τ) and their AUC values expressed as $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$. In the experiments, we use $\text{AUC}_{(F,\tau)}$ and AUC_{BS} [82] to represent the BS rate, and $\text{AUC}_{(D,F)}$ is used to evaluate the performance of the detector. AUC_{BS} can be computed by $\text{AUC}_{(D,F)}$ and $\text{AUC}_{(F,\tau)}$

$$\text{AUC}_{\text{BS}} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)}. \tag{33}$$

The accuracy (ACC) as the average test $\text{AUC}_{(D,F)}$ of all tasks is utilized to better assess the capacity of CL to identify anomalies. Meanwhile, backward transfer (BWT) represents the average forgetting measure of the model after completing T tasks. In order to explore the learning plasticity of new tasks in OHAD, we also use the forward transfer (FWT) to continuously estimate the learning ability for the new task. The following are the formulas of ACC, BWT, and FWT:

$$\text{ACC} = \frac{1}{T}\sum_{i=1}^{T} \text{AUC}_{(D,F)T,i} \tag{34}$$

$$\text{BWT} = \frac{1}{T-1}\sum_{i=1}^{T-1} \left(\text{AUC}_{(D,F)T,i} - \text{AUC}_{(D,F)T-1,i}\right) \tag{35}$$

$$\text{FWT} = \frac{1}{T-1}\sum_{i=2}^{T} \left(\text{AUC}_{(D,F)T,i} - \text{AUC}^{*}{}_{(D,F)i}\right). \tag{36}$$

$\text{AUC}^{*}{}_{(D,F)i}$ represents the result of training on the ith task. The $\text{AUC}_{(D,F)T,i}$ represents the ith task after learning the Tth task, where T represents the total number of tasks, and i refers to the ith task being evaluated. During the inference procedure, we evaluate the model after training on all tasks.

4) Number of N: For the sample replay strategy, the number of samples in each cluster is $P = 1$, $P = 2$, and $P = 3$, and the specific value of the total replay samples N is selected adaptively, which is shown in Table III.

TABLE III
NUMBER OF SAMPLES IN EACH CLUSTER (TOTAL NUMBER AND SELECTED NUMBER) AND THE SPECIFIC VALUE OF N IN THE SAMPLE REPLAY

HSIs	P=1		P=2		P=3		N
	Total	Selected	Total	Selected	Total	Selected	
Los Angeles-1	286	2	4603	46	5111	51	99
Bay Champagne	6603	66	2272	22	1125	11	99
Los Angeles-2	4919	49	2263	22	2818	28	99
Cat Island	364	3	3531	35	18605	186	224
San Diego	529	5	8858	88	613	6	99
HAD100_91	1734	17	886	8	1476	14	39
HAD100_100	1243	12	676	6	2177	21	39
HAD100_24	769	7	952	9	2375	23	39
HAD100_33	161	1	2432	24	1503	15	40
HAD100_40	2348	23	158	1	1590	15	39

B. Main Results

To verify the accuracy improvements and the robustness of our proposed CL-BioGAN for OHAD tasks under open scenario circumstances, we conducted comprehensive evaluations of the aforementioned six comparable methods on five OHAD datasets. CL-CaGAN is a comparable SOTA method for unsupervised continual OHAD. MAS and EWC are two representative strategies for regular, continuous learning method for traditional RGB images, respectively. We change the application form to the OHAD task to verify the effectiveness. Intuitively, these two methods appear to be better suited for the CL scenario. Furthermore, OWM is a representative method of gradient orthogonality in the CL family. Therefore, we mainly use these CL-based algorithm to evaluate and demonstrate the effect and performance in this article.

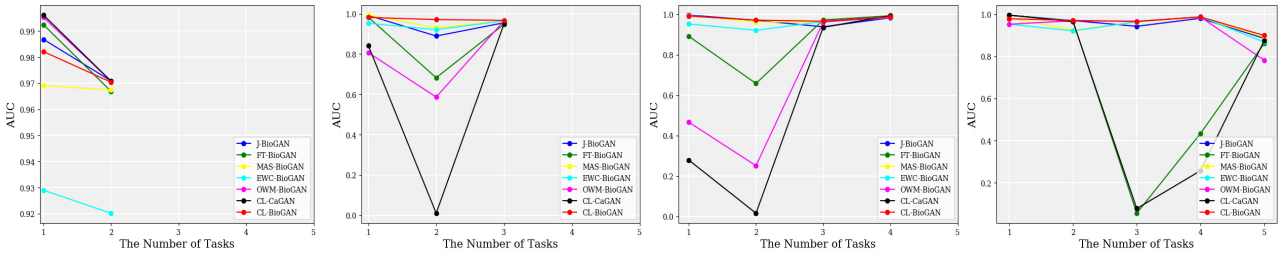
1) Quantitative Analysis: As shown in Table IV, we evaluate the performance of these methods for OHAD tasks, and the experiments cover different task combinations from tasks 2 to tasks 5. For instance, task 2 means the CL-based OHAD for Los Angeles-1 and Bay Champagne, task 5 means the CL-based OHAD from Los Angeles-1 to San Diego. We mainly analyze CL-based strategies and the backbone effect between BioGAN and CaGAN. The results in Table IV indicate that our proposed method CL-BioGAN presents the best ACC for most of the OHAD tasks, i.e., tasks 3–5. Specifically, this result not only displays the high ACC of CL-BioGAN in processing continuous tasks, but also indicates the satisfying stability and robustness ability to avoid forgetting old tasks when learning new tasks. Parameter comparison further demonstrates that our method achieves high accuracy with fewer parameters while maintaining robust accuracy improvement for learning new tasks. The testing results of catastrophic forgetting performance on previous tasks are shown in Fig. 4, which demonstrates that CL-BioGAN is capable of retaining important knowledge from previous tasks while learning new tasks, thereby presenting more stable AUC for each task. From ACC, BWT and FWT performance presented in Fig. 5, we can observe that our proposed CL-BioGAN indicates more robustness and reliability for all the tasks 5 in terms of involved transformer structure and biological-inspired AF Loss, and exhibits potential practical applications for the proposed CL strategy. Fig. 6 further demonstrates that our method exhibits more stable detection capabilities across different AUC curves. In addition, the BS rates are illustrated in Table V as new tasks are learned, where the larger the BS rate means the better the discrepancy between anomalies and backgrounds.

For the HAD100 dataset, we compared the CL-CaGAN and CL-BioGAN methods, as shown in Table VI. The ACC, BWT, FWT, and Avg BS are valued after continual anomaly detection on the five HAD100 datasets. The experiments cover different tasks combinations from tasks 2 to tasks 5. For instance, tasks 2 means the CL-based OHAD for HAD100_91 and HAD100_100, tasks 5 means the CL-based OHAD from HAD100_91 to HAD100_40. It can be clearly observed that CL-BioGAN achieves a better balance between memory stability and plasticity, resulting in less forgetting. Meanwhile, the Average BS values further reinforce the conclusion that CL-BioGAN exhibits strong stability in continual anomaly detection.

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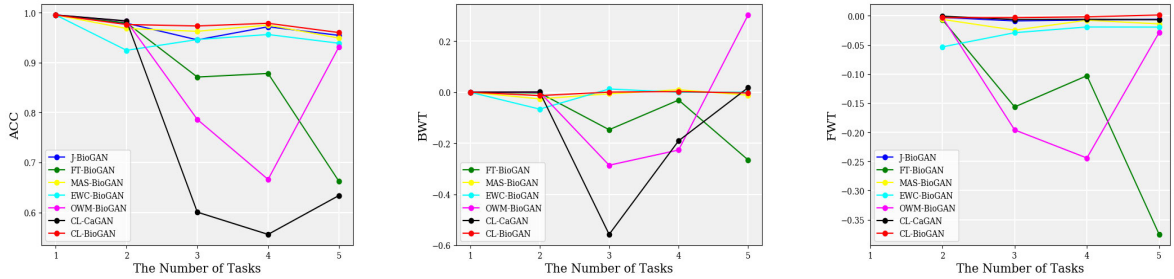
TABLE IV
CL FOR OHAD PERFORMANCE WITH THREE EVALUATION METRICS

ABU Dataset Tasks	Metrics	Method J-BioGAN	FT-BioGAN	MAS-BioGAN	EWC-BioGAN	OWM-BioGAN	CL-CaGAN	CL-BioGAN(ours)
2	ACC↑	0.9789	0.9797	0.9682	0.9246	0.983	**0.9835**	0.9763
	BWT↑	-	-0.0030	-0.0263	-0.0664	-0.0002	**0.0008**	-0.0133
	FWT↑	-	-0.0067	-0.0062	-0.0534	**-0.0028**	-0.0070	-0.0032
3	ACC↑	0.9456	0.8711	0.9627	0.9462	0.7862	0.6005	**0.9734**
	BWT↑	-	-0.1471	-0.0062	**0.0125**	-0.2863	-0.5575	0.0003
	FWT↑	-	-0.1565	-0.0246	-0.0290	-0.1960	-0.0064	**-0.0026**
4	ACC↑	0.9631	0.8783	0.9753	0.9563	0.6660	0.5559	**0.9788**
	BWT↑	-	-0.0309	**0.0096**	-0.0005	-0.2267	-0.1906	0.0025
	FWT↑	-	-0.1029	-0.0076	-0.0192	-0.2442	-0.0061	**-0.0022**
5	ACC↑	0.9497	0.6625	0.9491	0.9388	0.9313	0.6337	**0.9602**
	BWT↑	-	-0.2655	-0.0117	0.0002	**0.3026**	0.0178	-0.0033
	FWT↑	-	-0.3754	-0.0139	-0.0194	-0.0286	-0.0067	**0.0010**
-	Training Parameters↓	-	-	-	-	-	1140426	**752208**
-	Detection Parameters↓	-	-	-	-	-	921578	**628920**



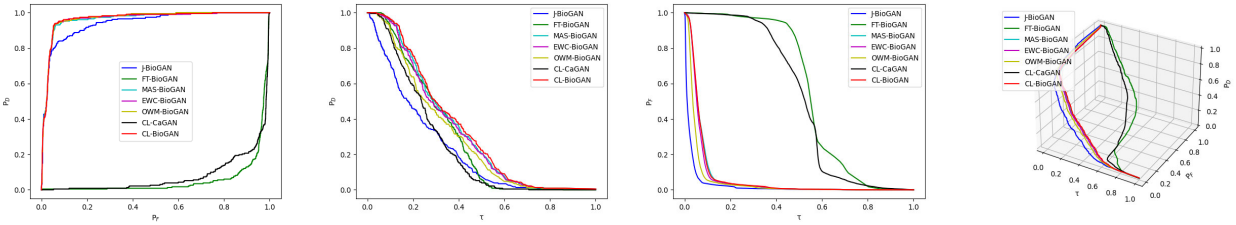
(a) (b) (c) (d)

Fig. 4. Catastrophic forgetting performance evaluation for open scenario OHAD on previous tasks. (a) AUC value for tasks 1 and 2 after training on the second task. (b) AUC value for tasks 1–3 after training on the third task. (c) AUC value for tasks 1–4 after training on the fourth task. (d) AUC value for tasks 1–5 after training on the fifth task.



(a) (b) (c)

Fig. 5. ACC, BWT, and FWT Performance evaluation for OHAD of different methods. (a) Variation of ACC for tasks 1–5. (b) Variation of BWT for OHAD tasks 1–5. (c) Variation of FWT for OHAD tasks 1–5.

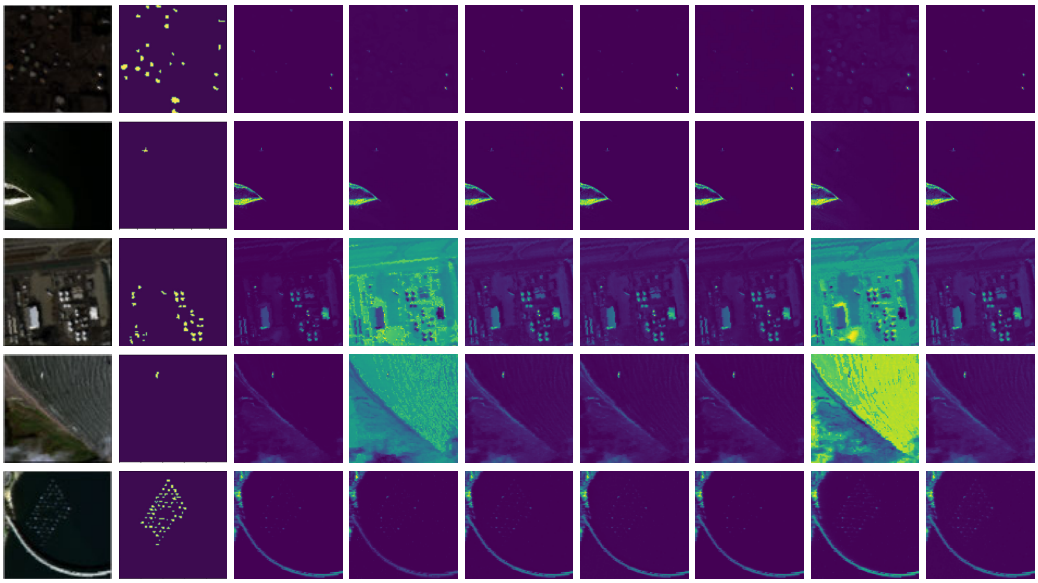


(a) (b) (c) (d)

Fig. 6. AUC testing results of different methods on the previous task after learning the fifth task are shown as follows. The first column (a) represents the $\mathrm{AUC}_{(D,F)}$ value on Los Angeles-2 after training tasks 1–5. The second column (b) represents the $\mathrm{AUC}_{(D,\tau)}$ value on Los Angeles-2 after training tasks 1–5. The third column (c) represents the $\mathrm{AUC}_{(F,\tau)}$ value on Los Angeles-2 after training tasks 1–5. (d) 3-D ROC value on Los Angeles-2 for tasks 1–5 after completing training on the fifth task.

2) Visual Detection Results: As illustrated in Fig. 7, CL-BioGAN demonstrates more stable detection capacity for the previous detection scenes/tasks after learning the knowledge of multiple scenes/tasks, the detection performance does not affected by surrounding noisy pixels. Compared to CL-CaGAN, we can observe from Fig. 7(h) and (i) that our method exhibits more robust detection performance for different anomalies. Meanwhile, CL-BioGAN can minimize

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 7. Visualization for five OHAD results after learning tasks 1–5. The dataset from top to bottom is Los Angeles-1, Cat Island, Los Angeles-2, San Diego, and Bay Champagne, respectively. (a) False color image of HAD. (b) Ground-truth map. (c) J-BioGAN. (d) FT-BioGAN. (e) MAS-BioGAN. (f) EWC-BioGAN. (g) OWM-BioGAN (h) CL-CaGAN. (i) CL-BioGAN.

TABLE V

AUC_{BS} IMPACT ANALYSES OF DIFFERENT CL STRATEGIES. BS REPRESENT THE VALUE OF AUC_{BS}, THE LARGER VALUE MEANS THE BETTER PERFORMANCE. AVG REPRESENTS THE MEAN VALUE OF BS. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD

ABU Dataset Tasks / Methods	1 $BS\uparrow$	2 $BS\uparrow$	3 $BS\uparrow$	4 $BS\uparrow$	5 $BS\uparrow$	Avg↑
J-BioGAN	0.9938	-	-	-	-	0.9938
FT-BioGAN	**0.9952**	-	-	-	-	**0.9952**
MAS-BioGAN	**0.9952**	-	-	-	-	**0.9952**
EWC-BioGAN	**0.9952**	-	-	-	-	**0.9952**
OWM-BioGAN	**0.9952**	-	-	-	-	**0.9952**
CL-CaGAN	0.9951	-	-	-	-	0.9951
CL-BioGAN	**0.9952**	-	-	-	-	**0.9952**
J-BioGAN	0.9853	0.9571	-	-	-	0.9712
FT-BioGAN	0.9873	0.9535	-	-	-	0.9704
MAS-BioGAN	0.9682	0.9517	-	-	-	0.9599
EWC-BioGAN	0.9260	0.9007	-	-	-	0.9134
OWM-BioGAN	0.9936	**0.9574**	-	-	-	0.9755
CL-CaGAN	**0.9953**	0.9571	-	-	-	**0.9762**
CL-BioGAN	0.9805	0.9530	-	-	-	0.9668
J-BioGAN	0.9790	0.8680	**0.9280**	-	-	0.9250
FT-BioGAN	0.9513	0.6326	0.9162	-	-	0.8334
MAS-BioGAN	**0.9921**	0.9150	0.9021	-	-	0.9364
EWC-BioGAN	0.9514	0.9042	0.9001	-	-	0.9186
OWM-BioGAN	0.7959	0.5564	0.9225	-	-	0.7583
CL-CaGAN	-0.0932	-0.6166	0.8341	-	-	0.0414
CL-BioGAN	0.9803	**0.9544**	0.9018	-	-	**0.9455**
J-BioGAN	**0.9949**	**0.9576**	0.8929	0.9378	-	**0.9458**
FT-BioGAN	0.0691	-0.0260	0.9375	0.9840	-	0.4912
MAS-BioGAN	0.9907	0.9447	0.8857	0.9304	-	0.9379
EWC-BioGAN	0.9505	0.9033	0.8986	0.9438	-	0.9241
OWM-BioGAN	0.4403	0.1755	**0.9313**	0.9671	-	0.6286
CL-CaGAN	-0.7180	-0.9379	0.7055	**0.9694**	-	0.0048
CL-BioGAN	0.9889	0.9534	0.8943	0.9328	-	0.9424
J-BioGAN	**0.9954**	**0.9573**	0.8957	**0.9634**	0.8553	**0.9334**
FT-BioGAN	0.9827	0.9467	-0.5055	-0.0639	0.8444	0.4409
MAS-BioGAN	0.9809	0.9063	0.8820	0.9373	0.8613	0.9136
EWC-BioGAN	0.9504	0.9027	**0.8971**	0.9446	0.8356	0.9061
OWM-BioGAN	0.9487	0.9545	0.916	0.9553	0.7586	0.9066
CL-CaGAN	0.9784	0.9390	-0.4341	-0.4201	0.8347	0.3796
CL-BioGAN	0.9864	0.9526	0.8942	0.9304	**0.8625**	0.9232

the false positives of anomalies in terms of the local-global attention mechanism involved in the transformer structure with a designed AF Loss for efficient BS.

TABLE VI

CL FOR OHAD PERFORMANCE WITH THREE EVALUATION METRICS AND AVERAGE AUC_{BS} IN HAD100 DATASET. AVG BS REPRESENT THE VALUE OF AVERAGE AUC_{BS}

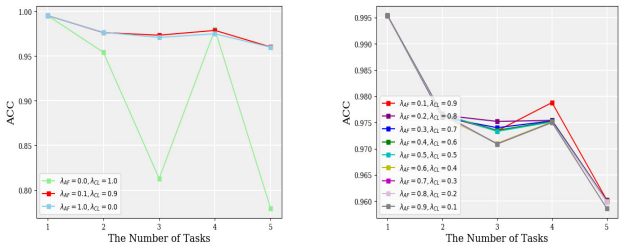
HAD100 Dataset Tasks	Metrics	Method: CL-CaGAN	Method: CL-BioGAN(ours)
2	ACC↑	0.9740	**0.9815**
	BWT↑	-0.0289	**-0.0129**
	FWT↑	-0.0220	**0.0022**
	Avg BS↑	**0.9546**	0.9441
3	ACC↑	0.9213	**0.9833**
	BWT↑	-0.0861	**0.0004**
	FWT↑	-0.1022	**-0.0041**
	Avg BS↑	0.8940	**0.9420**
4	ACC↑	0.9418	**0.9877**
	BWT↑	**0.0014**	0.0003
	FWT↑	-0.0629	**-0.0026**
	Avg BS↑	0.8821	**0.9305**
5	ACC↑	0.9595	**0.9885**
	BWT↑	**0.0101**	0.0001
	FWT↑	-0.0411	**-0.0021**
	Avg BS↑	0.8695	**0.9332**

C. Ablation Experiment

1) Effect Analysis of λ_{AF} and λ_{CL}: A detailed effect and assessment of CL performance for the proportions of λ_{AF} and λ_{CL} is illustrated in Fig. 8. We can observe that the network stability is enhanced with the addition of AF Loss, which helps to learn new knowledge more effectively when new tasks arrive. The results proves that the proposed AF Loss and CL loss can efficiently alleviate the issue of catastrophic forgetting in CL. We selected the optimal λ_{AF} and λ_{CL} value for the following experiments.

2) Effect Analysis for BSM: The comparison between the number of selected pixels and the real background pixels is shown in Table VII. We can note that background selection based on the SAM strategy indicates more accurate background pixel selection in most cases, which also provides the basic information for the generator to accurately learn reconstruction features of the background pixel.

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(a) (b)

Fig. 8. Varied parameter proportions of λ_{AF} and λ_{CL} for the detection ACC. (a) Individual impact of λ_{AF} and λ_{CL} on the experimental results and (b) impact of λ_{AF} and λ_{CL} on the experimental results within a certain range of values.

TABLE VII

RESULT OF IDENTICAL PIXELS TO THE TOTAL NUMBER OF PIXELS (RATIO1) AND CORRECTLY IDENTIFIED BACKGROUND PIXELS TO THE ACTUAL BACKGROUND PIXEL COUNT (RATIO2) FOR EUCLIDEAN DISTANCE (ED) AND SAM IN ROUND PIXELS IN GROUND TRUTH

HSIs	ED		SAM		GT
	ratio1	ratio2	ratio1	ratio2	
Los Angeles-1	0.6832	0.6798	**0.9739**	**0.9979**	9728
Cat Island	0.8782	0.8781	**0.9893**	**0.9898**	22481
Los Angeles-2	0.0723	0.0503	**0.9762**	**0.9991**	9768
San Diego	0.0202	0.0196	**0.9990**	**0.9993**	9989
Bay Champagne	0.6142	0.6062	**0.9571**	**0.9758**	9798

TABLE VIII

ANALYZING THE IMPACT OF THE NUMBER OF CLUSTERING GROUPS FOR CL-BIOGAN. AUC MEANS THE VALUE OF $AUC_{(D,F)}$. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD

ABU Dataset Tasks P	1 AUC↑	2 AUC↑	3 AUC↑	4 AUC↑	5 AUC↑	ACC↑	BWT↑
$P=2$	0.9954	-	-	-	-	0.9954	-
$P=3$	0.9954	-	-	-	-	0.9954	-
$P=4$	0.9954	-	-	-	-	0.9954	-
$P=5$	0.9954	-	-	-	-	0.9954	-
$P=6$	0.9954	-	-	-	-	0.9954	-
$P=2$	0.9629	**0.9713**	-	-	-	0.9671	-0.0325
$P=3$	**0.9821**	0.9704	-	-	-	**0.9763**	**-0.0133**
$P=4$	0.9635	**0.9713**	-	-	-	0.9674	-0.0319
$P=5$	0.9632	**0.9713**	-	-	-	0.9673	-0.0322
$P=6$	0.9640	0.9712	-	-	-	0.9676	-0.0314
$P=2$	0.9708	**0.9716**	0.9684	-	-	0.9703	**0.0041**
$P=3$	**0.9817**	0.9713	0.9672	-	-	0.9734	0.0003
$P=4$	0.9709	**0.9716**	0.9684	-	-	0.9703	0.0039
$P=5$	0.9707	**0.9716**	0.9684	-	-	0.9702	0.0039
$P=6$	0.9664	0.9708	**0.9835**	-	-	**0.9736**	0.0010
$P=2$	0.9908	**0.9710**	**0.9658**	**0.9876**	-	**0.9788**	**0.0056**
$P=3$	**0.9911**	**0.9710**	0.9656	0.9875	-	**0.9788**	0.0025
$P=4$	0.9907	**0.9710**	0.9656	0.9875	-	0.9787	0.0055
$P=5$	0.9904	0.9709	0.9656	0.9875	-	0.9786	0.0054
$P=6$	0.9880	0.9709	0.9656	**0.9876**	-	0.9780	0.0013
$P=2$	0.9753	0.9705	**0.9659**	**0.9876**	0.9005	0.9600	-0.0040
$P=3$	0.9786	0.9701	**0.9659**	0.9874	0.8992	0.9602	-0.0033
$P=4$	0.9742	**0.9706**	0.9658	**0.9876**	**0.9007**	0.9598	-0.0042
$P=5$	**0.9865**	0.9702	0.9656	0.9874	0.8995	**0.9618**	**-0.0012**
$P=6$	0.9805	0.9702	**0.9659**	0.9874	0.8994	0.9607	-0.0020

3) Effect Analysis of the Number of Clustering Group: Table VIII provides an insightful analysis of the impact of varying cluster numbers P in the K-means algorithm for CL OHAD task performance within the sample replay strategy framework. Across the range of cluster numbers studied from $P=2$ to $P=6$, we can notice that the configuration with $P=3$ consistently yields the highest accuracy. While configurations with $P=5$ and $P=6$ indicate a slightly increasing BWT in certain tasks, $P=3$ provides a more consistent performance across all task configurations. Therefore, the choice of $P=3$ represents an optimal balance between ACC and BWT. Although other P values exhibit minor advantages in individual metrics, $P=3$ delivers the optimal or suboptimal performance across all principal indicators.

TABLE IX

ACC AND BWT IMPACT ANALYSES FOR DIFFERENT COMPONENTS OF THE PROPOSED CL-BIOGAN. W/O R MEANS EXCLUDE OF REPLAY, W/O T MEANS EXCLUDE OF $\mathbf{L}_2$ SA, W/O L_{AF} MEANS EXCLUDE OF AF LOSS, W/O L_{CL} MEANS EXCLUDE OF CL LOSS, AND OURS MEANS CL-BIOGAN. AUC REPRESENT THE VALUE OF $AUC_{(D,F)}$. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD

ABU Tasks method	1 AUC↑	2 AUC↑	3 AUC↑	4 AUC↑	5 AUC↑	ACC↑	BWT↑
w/o R	0.9954	-	-	-	-	0.9954	-
w/o T	**0.9963**	-	-	-	-	**0.9963**	-
w/o L_{AF}	0.9954	-	-	-	-	0.9954	-
w/o L_{CL}	0.9954	-	-	-	-	0.9954	-
ours	0.9954	-	-	-	-	0.9954	-
w/o R	0.9827	0.9703	-	-	-	0.9765	-0.0127
w/o T	0.9834	0.9693	-	-	-	0.9764	-0.0129
w/o L_{AF}	0.9830	0.9702	-	-	-	**0.9766**	-0.0124
w/o L_{CL}	**0.9884**	0.9200	-	-	-	0.9542	**-0.0070**
ours	0.9821	**0.9704**	-	-	-	0.9763	-0.0133
w/o R	0.9767	0.9700	0.9659	-	-	0.9709	-0.0032
w/o T	0.9814	0.9695	0.9658	-	-	0.9722	-0.0009
w/o L_{AF}	0.9767	0.9700	0.9659	-	-	0.9709	-0.0033
w/o L_{CL}	0.5594	0.9207	0.9590	-	-	0.8130	-0.2142
ours	**0.9817**	**0.9713**	**0.9672**	-	-	**0.9734**	**0.0003**
w/o R	0.9767	0.9700	0.9659	0.9874	-	0.9750	0.0000
w/o T	0.9804	0.9700	0.9658	0.9873	-	0.9759	-0.0002
w/o L_{AF}	0.9767	0.9700	0.9659	0.9874	-	0.9750	0.0000
w/o L_{CL}	**0.9961**	0.9685	**0.9668**	0.9874	-	**0.9797**	**0.1641**
ours	0.9911	**0.9710**	0.9656	**0.9875**	-	0.9788	0.0025
w/o R	0.9767	0.9700	0.9659	0.9874	**0.8998**	0.9599	**0.0000**
w/o T	0.9256	**0.9705**	**0.9669**	**0.9875**	0.8973	0.9496	-0.0133
w/o L_{AF}	0.9767	0.9700	0.9659	0.9874	**0.8998**	0.9599	**0.0000**
w/o L_{CL}	0.9331	0.1149	0.9676	**0.9875**	0.8944	0.7795	-0.2289
ours	**0.9786**	0.9701	0.9659	0.9874	0.8992	**0.9602**	-0.0033

4) Other Ablation Effect Experiment: We mainly evaluate and analyze the impact of the $\mathbf{L}_2$ SA structure in D, the replay strategy, the AF Loss, and the CL loss for OHAD performance across five different scenario tasks. The effect of above mentioned components utilized in our proposed CL-BioGAN is illustrated in Table IX, respectively. We can observe that CL-BioGAN with all above mentioned components presents the optimal ACC and BWT results in most cases especially after CL from the second task. For the modal of CL-BioGAN without AF Loss, it exhibits unstable changes in ACC and significant fluctuations in BWT when learning new scenario tasks. In contrast, CL-BioGAN demonstrates more stable learning of new tasks while effectively preventing catastrophic forgetting issues. Particularly, after learning the third scenario, the CL-BioGAN demonstrates beneficial retention of knowledge for previous scenarios, which indicates that AF Loss enhances sustained detection performance while increasing the robustness and stability of the model.

Table X reflects the AUC_{BS} impact analyses for different components of the proposed CL-BioGAN. As new tasks are learned, AF Loss may degrade the BS performance in certain scenarios, while AF Loss ensures the stability of BS. While for the CL-BioGAN Without AF Loss, it exhibits significant fluctuations as new knowledge is incorporated. This performance

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TABLE X

AUC_{BS} IMPACT ANALYSES FOR DIFFERENT COMPONENTS OF CL-BIOGAN. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD

ABU Tasks / Methods	1 $BS\uparrow$	2 $BS\uparrow$	3 $BS\uparrow$	4 $BS\uparrow$	5 $BS\uparrow$	Avg↑
w/o R	**0.9952**	-	-	-	-	**0.9952**
w/o T	0.9951	-	-	-	-	0.9951
w/o L_{AF}	**0.9952**	-	-	-	-	**0.9952**
w/o L_{CL}	**0.9952**	-	-	-	-	**0.9952**
ours	**0.9952**	-	-	-	-	**0.9952**
w/o R	0.9808	0.9526	-	-	-	0.9667
w/o T	0.9820	0.9520	-	-	-	0.9670
w/o L_{AF}	0.9810	**0.9545**	-	-	-	**0.9697**
w/o L_{CL}	**0.9830**	0.8906	-	-	-	0.9368
ours	0.9805	0.9530	-	-	-	0.9668
w/o R	0.9744	0.9523	0.8933	-	-	0.9400
w/o T	0.9800	0.9526	0.8970	-	-	0.9432
w/o L_{AF}	0.9744	0.9523	0.8933	-	-	0.9400
w/o L_{CL}	0.5553	0.8987	0.8762	-	-	0.7767
ours	**0.9803**	**0.9544**	**0.9018**	-	-	**0.9455**
w/o R	0.9743	0.9522	0.8933	0.9295	-	0.9373
w/o T	0.9785	0.9528	0.8955	0.9318	-	0.9397
w/o L_{AF}	0.9743	0.9522	0.8933	0.9295	-	0.9373
w/o L_{CL}	**0.9954**	**0.9541**	**0.9135**	**0.9511**	-	**0.9535**
ours	0.9889	0.9534	0.8943	0.9328	-	0.9424
w/o R	0.9742	0.9522	0.8932	0.9294	0.8620	0.9222
w/o T	0.9231	0.9520	0.8949	0.9163	0.8564	0.9085
w/o L_{AF}	0.9742	0.9522	0.8932	0.9294	0.8620	0.9222
w/o L_{CL}	0.9287	-0.0023	**0.9076**	**0.9421**	**0.8678**	0.7288
ours	**0.9764**	**0.9526**	0.8942	0.9304	0.8625	**0.9232**

indicates that AF Loss is a key component for sustaining CL ability and serves as a stabilizer for BS. The similar results between $AUC_{(D,F)}$ in Table IX and AUC_{BS} in Table X indicate that CL-BioGAN reduces false positives in $AUC_{(D,F)}$ as well as enhances the AUC_{BS} performance, which further demonstrates the proposed CL-BioGAN can facilitate better generation for pseudobackground pixels in OHAD.

V. CONCLUSION

In this study, in order to mitigate catastrophic forgetting as well as adapt to new endless tasks coming in the cross-domain OHAD, a novel CL-BioGAN is proposed to improve the detection accuracy and robustness through a designed bioinspired AF Loss and SA-based GAN structure. The proposed CL-BioGAN incorporates a continuous replay strategy by CL-Bio loss to preserve historical knowledge and adapt to new tasks in an open scenario circumstance. AF Loss is designed to balance stability and plasticity by modulating the forgetting rate of previous knowledge, which is functionally consistent with the advantage of biological forgetting in memory flexibility. This connection provides a underlying mechanism theoretical explanation for the conflict between biological active forgetting and continually learning experiences. In addition, CL-BioGAN achieves an end-to-end reconstruction by the synergistic integration of a generator and an L_2 SA discriminator within a GAN framework for effectively learning representative spectral characteristics of background distribution through differentiable data augmentation. Our experiments on five cross-domain OHAD datasets indicate that CL-BioGAN significantly offers more stable OHAD performance while maintaining a balance between historical and new tasks.

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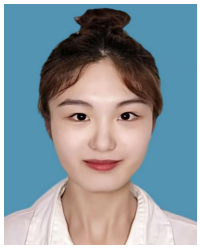
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